
Step 1: Understanding a 2D Array

A **2D array** is like a table with rows and columns. Suppose we have this array:

```
int[][] numbers = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}  
};
```

This means:

```
Row 0 → [1, 2, 3]  
Row 1 → [4, 5, 6]  
Row 2 → [7, 8, 9]
```

Step 2: Writing the Loop

To go through all elements, we use **two loops**:

1. **Outer loop** for rows (*i*).
2. **Inner loop** for columns (*j*).

```
for (int i = 0; i < numbers.length; i++) {           // Outer loop for rows  
    for (int j = 0; j < numbers[i].length; j++) {    // Inner loop for columns  
        System.out.print(numbers[i][j] + " ");  
    }  
    System.out.println(); // Move to a new line after printing a row  
}
```

Step 3: Step-by-Step Execution

Let's execute this code **line by line**.

First Outer Loop (*i* = 0, First Row)

- *i* = 0, so we're in **Row 0** → {1, 2, 3}
- **Inner loop runs (*j* varies from 0 to 2):**
 - *j* = 0 → numbers[0][0] = 1 → Prints 1
 - *j* = 1 → numbers[0][1] = 2 → Prints 2
 - *j* = 2 → numbers[0][2] = 3 → Prints 3
- Inner loop ends → System.out.println(); moves to the **next line**.

Second Outer Loop (i = 1, Second Row)

- $i = 1$, so we're in **Row 1** $\rightarrow \{4, 5, 6\}$
- **Inner loop runs (j varies from 0 to 2):**
 - $j = 0 \rightarrow \text{numbers}[1][0] = 4 \rightarrow \text{Prints } 4$
 - $j = 1 \rightarrow \text{numbers}[1][1] = 5 \rightarrow \text{Prints } 5$
 - $j = 2 \rightarrow \text{numbers}[1][2] = 6 \rightarrow \text{Prints } 6$
- Inner loop ends $\rightarrow \text{System.out.println()};$ moves to the **next line**.

Third Outer Loop (i = 2, Third Row)

- $i = 2$, so we're in **Row 2** $\rightarrow \{7, 8, 9\}$
 - **Inner loop runs (j varies from 0 to 2):**
 - $j = 0 \rightarrow \text{numbers}[2][0] = 7 \rightarrow \text{Prints } 7$
 - $j = 1 \rightarrow \text{numbers}[2][1] = 8 \rightarrow \text{Prints } 8$
 - $j = 2 \rightarrow \text{numbers}[2][2] = 9 \rightarrow \text{Prints } 9$
 - Inner loop ends $\rightarrow \text{System.out.println()};$ moves to the **next line**.
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Final Output

After all iterations, the **printed output** will be:

```
1 2 3
4 5 6
7 8 9
```

Summary

1. **Outer loop** controls the rows (i).
2. **Inner loop** controls the columns (j).
3. Elements are accessed using `numbers[i][j]`.
4. `System.out.println();` moves to a new line after each row.